

Logistic Regression Part II

We have k classes. In the training set for a particular example X , it belongs to only one class Y_c . So $y_c = 1$ and $Y_j = 0$ for all j not equal to c . Y looks like $[0,0,0,..1, 0,0,0]$. This is called one hot vector.

The multinomial logistic classifier uses a generalization of the sigmoid, called the **softmax** function, to compute $p(y_k = 1|\mathbf{x})$. The softmax function takes a vector $\mathbf{z} = [z_1, z_2, \dots, z_K]$ of K arbitrary values and maps them to a probability distribution, with each value in the range $[0,1]$, and all the values summing to 1. Like the sigmoid, it is an exponential function.

For a vector $\mathbf{z}$ of dimensionality K , the softmax is defined as:

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)} \quad 1 \leq i \leq K \quad (5.16)$$

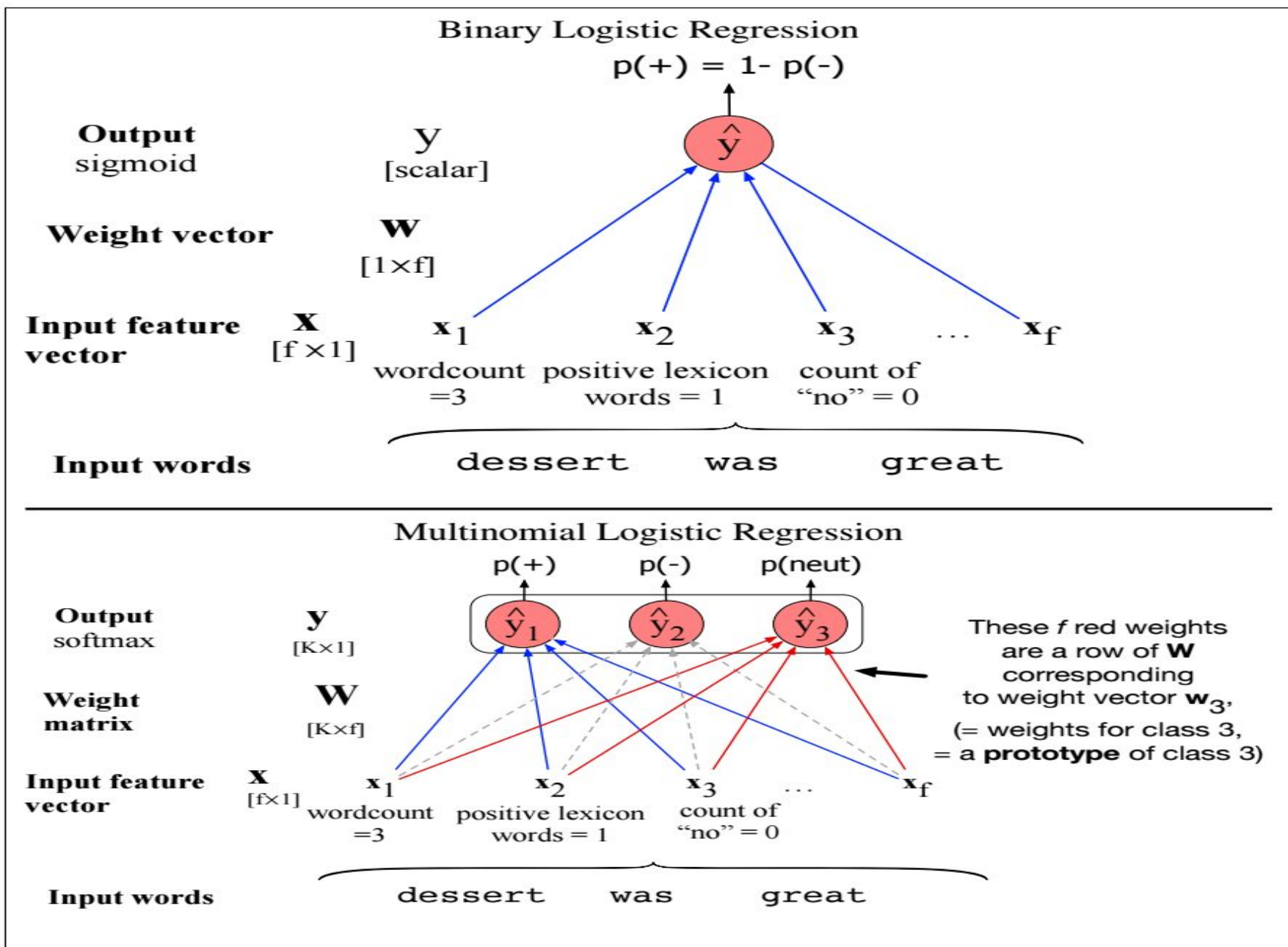
$$\text{softmax}(\mathbf{z}) = \left[\frac{\exp(z_1)}{\sum_{i=1}^K \exp(z_i)}, \frac{\exp(z_2)}{\sum_{i=1}^K \exp(z_i)}, \dots, \frac{\exp(z_K)}{\sum_{i=1}^K \exp(z_i)} \right]$$

$\mathbf{z} = [0.6, 1.1, -1.5, 1.2, 3.2, -1.1]$ After applying softmax function the probabilities of the classes are $[0.05, 0.09, 0.01, 0.1, 0.74, 0.01]$

$$p(y_k = 1 | \mathbf{x}) = \frac{\exp(\mathbf{w}_k \cdot \mathbf{x} + b_k)}{\sum_{j=1}^K \exp(\mathbf{w}_j \cdot \mathbf{x} + b_j)}$$

Each row k of W corresponds to the vector of weights $\mathbf{w}_k$. W thus has shape $[K \times f]$, for K the number of output classes and f the number of input features. The bias vector b has one value for each of the K output classes. If we represent the weights in this way, we can compute $\hat{y}$, the vector of output probabilities for each of the K classes, by a single matrix multiplication

$$\hat{y} = \text{softmax}(W\mathbf{x} + b)$$



$$L_{\text{CE}}(\hat{y}, y) = -\log p(y|x) = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

$$\begin{aligned} L_{\text{CE}}(\hat{\mathbf{y}}, \mathbf{y}) &= -\sum_{k=1}^K y_k \log \hat{y}_k \\ &= -\log \hat{y}_c, \quad (\text{where } c \text{ is the correct class}) \\ &= -\log \hat{p}(y_c = 1 | \mathbf{x}) \quad (\text{where } c \text{ is the correct class}) \\ &= -\log \frac{\exp(\mathbf{w}_c \cdot \mathbf{x} + b_c)}{\sum_{j=1}^K \exp(\mathbf{w}_j \cdot \mathbf{x} + b_j)} \quad (c \text{ is the correct class}) \end{aligned}$$

$$\begin{aligned} \frac{\partial L_{\text{CE}}}{\partial w_{k,i}} &= -(y_k - \hat{y}_k) x_i \\ &= -(y_k - p(y_k = 1 | x)) x_i \\ &= -\left(y_k - \frac{\exp(\mathbf{w}_k \cdot \mathbf{x} + b_k)}{\sum_{j=1}^K \exp(\mathbf{w}_j \cdot \mathbf{x} + b_j)} \right) x_i \end{aligned}$$

Stochastic vs Minibatch processing

$$\begin{aligned} Cost(\hat{y}, y) &= \frac{1}{m} \sum_{i=1}^m L_{CE}(\hat{y}^{(i)}, y^{(i)}) \\ &= -\frac{1}{m} \sum_{i=1}^m y^{(i)} \log \sigma(\mathbf{w} \cdot \mathbf{x}^{(i)} + b) + (1 - y^{(i)}) \log (1 - \sigma(\mathbf{w} \cdot \mathbf{x}^{(i)} + b)) \end{aligned} \quad (5.33)$$

The mini-batch gradient is the average of the individual gradients from Eq. 5.30:

$$\frac{\partial Cost(\hat{y}, y)}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m [\sigma(\mathbf{w} \cdot \mathbf{x}^{(i)} + b) - y^{(i)}] x_j^{(i)} \quad (5.34)$$

Instead of using the sum notation, we can more efficiently compute the gradient in its matrix form, following the vectorization we saw on page 7, where we have a matrix $\mathbf{X}$ of size $[m \times f]$ representing the m inputs in the batch, and a vector $\mathbf{y}$ of size $[m \times 1]$ representing the correct outputs:

$$\begin{aligned} \frac{\partial Cost(\hat{y}, y)}{\partial \mathbf{w}} &= \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y})^\top \mathbf{X} \\ &= \frac{1}{m} (\sigma(\mathbf{X}\mathbf{w} + \mathbf{b}) - \mathbf{y})^\top \mathbf{X} \end{aligned} \quad (5.35)$$